

HOW DOES A TREE GROW ?

OR

BOTANY FOR YOUNG AUSTRALIANS.

BY

JAMES BONWICK,

*Sub-Inspector of Denominational Schools, Victoria,
Author of "Geography of Australia and New Zealand,"
&c. &c.*



JAMES J. BLUNDELL & Co., MELBOURNE;
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PREFACE.

AT the request of several Teachers, I have commenced a Shilling Series of School Books, chiefly to be confined to subjects of Colonial History and Popular Sciences.

The form of dialogue has been adopted with the “Botany for Young Australians,” from a belief that the sympathies of our young friends will be excited on behalf of the juvenile questioner, and their interest thus maintained in the study of the sciences.

A dialogue upon Astronomy will shortly follow; being a conversation between a father and his son, coming out to Australia, from Old England.

JAMES BONWICK.

Melbourne, April 17, 1857.

HOW A TREE GROWS IN AUSTRALIA.

WILLIE was a fine rosy-faced boy of our southern colony. Though not eight years of age, he was as healthy and merry a lad as ever climbed up a Gum tree, picked up manna, or rode in a bullock dray.

His father had once occupied a good position in Old England; but the uncertainties and losses of business, and the constant struggle to uphold a respectable appearance with decreasing means, became so burdensome to his mind, that his spirits failed, and his energies sunk. His attention was directed to Australia, the land of mutton and corn, the home of health and plenty. Gathering up the wreck of the past, he left the country of taxation and paupers, and established himself on a small farm in Port Phillip.

The young hero of our story had been a year or two in the colony. It so happened he had a piece of land of his own, in which he proudly exhibited some rising cabbages, a few peas, and a flower or two. His father had given him a rose tree, which was the reigning beauty of the bed. It was upon the occasion of his parent's visit to the garden, that the following dialogue took place:—

Look, father, and see how my rose tree has grown.

Indeed it has, Willie. Can you tell me what has made it grow?

The sun and the rain, I suppose.

Do you remember, when we got tired of the old slab hut, and set about building this brick cottage, that you noticed it getting higher and higher every day!

Yes, that was because more bricks and wood were used.

Then, if your tree increases in size, there surely must be something added on continually: do you think the sun and rain do this?

Well, I never thought about it, father; but I should like to know why it does grow.

Can you tell me, Willie, what a plum pudding is made of?

Yes, that I can. There is the flour, the suet, the raisins, and the cold water. All these are mixed together.

Then let us see of what our rose tree is made.

I don't think it so easy to tell that as to reckon up the articles in a pudding.

Never mind, we will try. First, there is the stalk, or woody part. When you put a piece of stick in the fire, what becomes of it?

Oh, it smokes and blazes, and then nothing is left but some ashes.

What is it which burns away?

That I cannot tell.

It is the gaseous part which burns in a flame, like what you have seen come out of coal. But what do you call woody matter that will not blaze?

Charcoal, father. Then I understand now that wood is nothing but charcoal and the gases. What are these gases?

The gas which blazes so readily, my dear, is hydrogen: and it has a very strong smell too. The air we breathe is a mixture of two gases—oxygen and nitrogen. It is only the oxygen that we take into our lungs.

Well, that is curious.

I shall puzzle you more, Willie, when I tell you that water is nothing but a mixture of oxygen gas and hydrogen gas.

It certainly is funny that water, which puts out flame, should be partly composed of the burning gas.

You must also know, my lad, that hydrogen would not burn without oxygen. You blow air into a fire to give food for flame.

But however could the plants get hold of the gases, father?

Did it never strike you why God formed leaves?

It never did, except that I thought he did it to make the trees look pretty.

That is quite true, Willie. The good God loves beauty, and he has surrounded us with beauty of all kinds. But he made things for use as well

as to be looked at. The leaves absorb or suck in gases from air and water.

Then I suppose the veins like that we see in leaves conduct these gases away into the plant.

Quite right, my boy. Where now shall we get the charcoal, or carbon, as the learned men call it?

That I cannot find out at all.

You told me, Willie, that smoke came out of burning wood. What becomes of it?

When I was a very little boy, I thought it went up to form clouds; but now I know part of it turns into soot in the chimney, and that looks like our charcoal or carbon.

It really is. As to that which comes out of the chimney, it passes upward, and gets gradually mixed with the air. The little particles of carbon join the oxygen, and become a sort of gas called carbonic acid gas, which is absorbed into the plant.

How wonderful that the solid part of a tree should once have been floating about in the air!

Do you think the leaves of a plant to be the same as the stem?

Yes, I do; for when they are thrown in a fire, they smoke, blaze, and leave an ash like the wood does.

Just so. You know the smoke to be carbon passing into the air; but we must examine the ash a little more carefully. If you take some ash from the fireplace, and put it into hot water, the solid part will of course fall to the bottom.

Will no part mix with the water?

There will be something; for if we pour off the water, and allow it to evaporate in a dish, there will be found to be a sediment left, and that is potash or pearl ash.

I have heard of people in the bush doing that when they could not get soap, for they said that the potash got the dirt out of clothes.

It is a great pity that we in these colonies burn away so much wood in waste when clearing land, Willie, without thinking of making potash out of the ashes, for it fetches a good price.

Then there is potash in the plant. Has any thing else been found in the ash beside that and carbon?

Yes, my lad. Sulphur or brimstone, lime, soda, flint, ammonia, phosphorus, magnesia, and iron, are contained in trees.

But how could all these things get there?

Why, if we cannot find them in the air to be absorbed by the leaves, they must be in the soil or ground. Now, it so happens that those substances are to be found in different quantities in different places.

How do they get into the plant, father?

Simply by the little rootlets absorbing small particles of them, mixed with moisture.

But do all plants require the same amount of lime, potash, soda, and the others?

No, my dear. There are not two sorts of trees that feed upon the same materials in exactly the same proportions.

Is that the reason, then, why some land is so much better fitted to grow one plant than another?

The reason is, because the one soil has more of the right sort of food in it.

Now I see that if I wanted to grow a good crop of any thing, I must give it plenty of the food it likes best.

Yes, but not too much. For like as too much nice rich food is bad for children, so it is with vegetables: ground may be too rich, as well as too poor.

I have heard people say that it is not wise to grow the same thing in the same soil year after year: why is this?

Because it would gradually consume all the food there, and then it would starve, and look miserable.

Then my beautiful flower-bed will by-and-by cease to bring forth such a fine show as it has done this season.

Of course it will, unless you provide your plants with fresh food.

Fresh food, father; I do not understand you.

I mean, manure must be mixed with the soil.

How is manure food for plants?

Because it contains the materials they require. You throw wood ashes over the ground, and so add sulphur, potash, and carbon. Sea weed manure furnishes plenty of soda. Bone dust contains lime and phosphorus.

It is possible, then, to apply to the ground the amount of solid matter taken out of it by the plant, so that if my radish bed had some manure, it would be as good as it was before my crop came off.

That is perfectly correct, my boy.

But how is it that a gum-tree forest is kept up, for there must be a tremendous lot of lime, soda, flint, and the rest, removed from the soil?

Yes, but when the trees fall, they rot, and the solid parts return to the ground.

Oh, father, the remains are very small, compared to the living tree.

True, because the principal part of a plant consists of the gases, which fly off, and of carbon, which unites with the oxygen of the air.

How does God bring fresh carbon to the forest?

Several ways: smoke is one source, and the breath of animals another.

What has the breath to do with it?

Every time you respire, or breathe out, some carbonic acid comes out with air, and is carried into the atmosphere.

Why, father, you do not mean to say that my breath helps to make cabbages grow.

The carbon passing from your body may become a part of a cabbage, or gum tree, or a delicate tulip.

The next time Willie and his father were out together, the conversation again fell upon trees. The wonder of the boy had been strongly excited by the last lesson, and he had now lots of questions to ask. He knew enough to know that there must be a great deal more to learn. He had been told that trees fed the same as animals, and he felt sure that inside there must be some entrances for the food to reach parts needing supply. Then he sought to understand how the growing process was managed, and especially how seeds were formed, and how the plant sprang from them. Thus, question after question poured out from the boy's lips, without even a pause for a reply.

"Stop, stop, my man," said his father; "I am not like the Hindoo god with half-a-dozen pairs of ears, and half-a-dozen tongues. We will go now a little deeper into the subject; but we must take one thing at a time. What do you think of that gum tree yonder?"

That is a noble fellow. What a barrel he has got for splitting paling out of! And hasn't he got a fine top knot? Why, that must be almost as big as that Tasmanian tree you read about.

Oh, no; that one was 350 feet high, and was 104 feet round; while this is not above 100 feet high, and 30 round.

Well, then, that must be a monster surely. How curious to think it was once a tiny little thing that I could pull up with my finger! I say, father, how many cartloads of carbon this one must have got hold of! I fancy it has got gas enough to fill many a balloon. But how did it grow?

To answer that question, will give us some trouble, and take some time. First, tell me all the parts of the tree.

What I cannot see is the root; then comes the stem, then the branches, and then the leaves.

You forget the flower.

Flower! whoever heard tell of a gum flower? How funny the word sounds!

If there be no flower, how are you to get the seed?

I never thought of that. But flowers are always such pretty light things, that one would be sure to see them a long way off on a gum tree.

But if instead of having fine red leaves, my lad, the flower had none, and the other part was much the same colour as the leaves, do you think you would notice it so readily?

No, father. Won't I give a good look out for it after this; for I am sure none of our boys at school ever talk of gum flowers, though we often go to gather wattle blossoms.

To go on with our tree—we will take the root, and there is a Stringy Bark blown over in the last storm.

And a strong root it has, too. How the wind must have puffed to overcome the weight of all the gravel and clay resting on that lot of roots, especially as they held the gravel like so many fingers. So these are the suckers of moisture and food out of the soil.

Yes, but you do not see the real suckers. They are very small, and were broken off and left behind. Those are called *spongioles*, because they suck up like a sponge. They are situated at the ends of the small fibres of roots, and have their mouths always open.

Yet I don't see why the moisture rises. If I put my mouth into the rain cask, the water will not rush up into it.

Your mouth is too big. Supposing you put a lump of salt near a little water, so as to be touched, what will be the result?

The salt will gradually absorb the water, until there be none left.

After dipping the corner of a towel in the basin, and hanging it up by the opposite corner, does not the dampness run down when drying?

No, the moisture ascends and wets the dry part of the towel.

So you see, then, that the water in small particles can arise in the pores, or narrow openings of substances. This is because the sides attract it, and the process is called *capillary*, or hairlike, *attraction*. On this principle the moisture rises into the rootlets through the *spongioles*.

Is that all the work they do?

Sometimes these fibres reject things not wanted in the tree, and that often prove poisonous to its growth. Such important little workmen as the spongioles ought not to be disturbed in their labour, and this is what makes the difficulty in moving trees.

I know that most of those I move are sure to die. But gardeners are more lucky.

The reason is, because they do it at a proper time.

What, father, can we catch the spongioles asleep?

Not exactly; but they only live a year, and you must take up the tree between the time of the death of one set, and the production of another, which is the winter season. Even then you should get up as many fine roots as possible, for at the end of these the spongioles grow.

All roots are not alike, father.

That is true: some are creeping, or fibrous, or bulbous, or knobbed, or taprooted. Can you give me instances of each?

Let me see. The mint spreads underground by its creeping roots. The potatoe is knobbed, and the onion and tulip are bulbous. The grass is fibrous, and the parsnip has a tap root.”

So much for the root. Now let us look at the bark. I suppose you have noticed the difference of the bark of our forest trees.

I know that they shed their bark, though not in the same way. The stringy bark peels off in strings. The gum throws out fine long ribands, waving in the wind. The iron bark sheds its thick coat in great lumps.

But does the whole of the bark thus fall off?

Oh, no: it is only the rough, worn-out stuff. There is always bark left. It puts me in mind of *my* hand, that got so horny after sawing a whole day at a big tree; for days after the rough skin got peeling off as if it was not wanted.

Then you have more skins than one. You are just like a tree, for that has several coats to its bark. Which is the softer, the outer or inner bark?

The outer is hard, and the inner soft. But there is a fresh gum tree just cut down: that will show us the barks.—Yes, now I peel off the outside, there is a very soft, juicy stuff, a thing I feel—a soft coat of bark close to the ring of white-looking wood.

Mind, Willie, the outside is the *cuticle* or *epidermis*, having pores or openings through which moisture issues at one time, and is absorbed at another. These *Stomates* or openings are very small; in our Bush Pigfaces, that the Blacks eat, there are 70,000 *Stomates* to every square inch of skin.

Then the tree perspires in the same manner that we do: that is odd. But what is the middle pulpy bark, with its green colour and sticky feeling?

Botanists call that the *Chlorophyll* or colouring matter. The inner bark is the *liber*, which you see to be soft and fibrous like, being full of sap.

Why, this is something like our skin, as there is a watery stuff between our cuticle and the skin that has the blood in it. I have heard of people writing on bark: that must have been on the liber.

It was so. The English wrote on the bok or bark, and thus we have the word *book*; while with *liber* we connect the word *library*. But we shall return to the liber in another lesson.

Is the cuticle always naked?

No, some plants I have seen with hair over the skin, and others with something like scales.

Having talked of the bark, boy, we will turn to the stem. What are your notions about it?

My eyes and fingers tell me that, after I pass the liber of the bark, I reach the fresh-looking white ring of wood, and that after that the wood which is still in rings gets harder and drier toward the centre. But I thought there was pith in the middle of a tree: how is it there is none seen here?

Because these woody layers or rings have crushed it in time. I must say a little about this pith. You know it is soft enough; that is, because it is

composed of small cells filled with juices at first, and then with air.

Cells, father! what, like the bees' honey-comb!

Yes, indeed, and six-sided like those cells. The walls, however, are not made of wax, but of *cellular tissue*. This tissue is a fine membrane, or skin; or, rather, a transparent, colourless, film. Mucous threads connect the cells.

What curious things to be in a tree?

I shall astonish you more when I tell you that all your tree—root, bark, pith, wood, and leaf—every one of these is composed of a number of these little cells. Do you see those small spots in the wood, and these in this leaf in my hand?

I do, father.

They are thought to be the openings of the cells, through which various juices may pass. Some cells are round and regular, as in the leaf; while those of the wood, from being much crushed, are spindle shaped, like a tap root.

Would they not be round in the fruit, then?

They would, Willie, and are there filled with some nice secretions, that boys rather enjoy. In the pretty leaves of flower, they contain a coloured fluid. Some cells hold carbon, and others have crystals of flint, lime, iron, starch, &c. Sugar, gum, resins, turpentine, &c., are in cells.

That is where the scratching stuff comes from, which I have felt in the cutting grass and wheat straw. But are there cells in roots?

Yes. There are, also, long tubes in plants like our water pipes, which are open at each end, not closed like the cells. They are parallel to the line of the stem, and are extremely small. They may be called elongated or lengthened cells, and form what is called the *vascular system*, or system of vessels.

Is there any other sort of vessels?

Some are *spiral*, or twisted. The *tracheae* are such. Some are round the pith. It is supposed that these convey air from one part to another.

Are these spirals everywhere?

No, there are none in the bark nor root—nor have ferns any. Speaking of air tubes, I may tell you that aquatic or water plants have air cells divided from other parts by layers of cellular tissue.

What do the spirals want with air?

You have been told that oxygen of the air is as necessary to the life and nourishment of a tree as of a boy; sometimes the leaves do not absorb enough, and then the spirals get it, through the root, from the moistened soil.

I can now understand why so large a thing as a tree becomes so small when the really solid part is got together. If all the cells and vessels were jammed up tight, so that the air and juices were driven out, how thin our thirty-foot round gum tree would appear! But will you tell me if the regular sap runs through those vessels we have talked of?

It is considered that it runs rather between the interstices or openings between the cells.

Why do you say *considered* and *supposed*?

Simply because the things we talk of are so very minute or small that the most powerful microscope will scarcely exhibit them, and thus it is difficult to observe their nature and action.

Father, what are these silver lines through the solid wood, coming from the pith centre?

They are the *medullary* or *pith rays*, and reach the liber or inner bark. They convey nourishment. Through them there is also a communication between the root and the leaf.

I want to ask a question about the leaf. Does it throw off anything as well as absorb?

You have been told that it absorbs oxygen from the air, and also carbonic acid gas—that is, carbon and oxygen. This carbonic acid floating in the air

is absorbed in the day-time, though the leaf exhales, or breathes out, oxygen gas while the sun shines.

Does it not do so at night?

No; then the leaf throws out some carbonic acid gas undigested. There is this use of the oxygen being taken in by the leaf, that it acts beneficially upon the sap to thicken it, &c.

What is the true shape of the leaf?

The course of veins or branching vessels. The fleshy part of the leaf is of cellular tissue.

I have heard that our leaves differ from English ones: how are they different? I know they do not fall off in autumn.

English leaves are made to fall, as from the first they are nearly divided from the *petiole* or leaf stalk, the mark of division being to be seen in the budding leaf.

Have our Australian leaves that mark?

No, they have not. The petiole expands or becomes leafy, and the leaves, therefore, do not readily fall. There is yet another difference. While the glands or tumours for imbibing moisture are on the underside of the leaf in England, they are on both sides of it here. Instead, too, of being horizontal, our leaves are set in edgewise on their stems.

I often notice the great difference there is in leaves, father.

Yes, they differ in size, structure, direction of veins, situation, form, duration, kind of surface, &c.

But what have you to say of those plants which have no leaves? I could never see any in our sheoak nor cherry.

Quite true. Our *Casuarina* or oak family, and the *Exocarpus*, or cherry, have no leaves. They have, however, long hanging knotted twigs at the end of branches. They have been called the Pines of Australia.

Is light necessary to leaves as well as air?

It is. Sunlight acts chemically to produce changes in substances. It is believed necessary to enable the leaf to separate the carbon from the oxygen

in carbonic acid gas. Leaves love light, and will turn toward it. Some even follow the sun in its daily motion. In a dark place, plants soon fade, and the leaf turns white.

Well, father, we have got rid of the root of our gum tree, as well as its bark and leaves—what shall we take next? I seem to know a good deal, but I cannot yet make out how the tree grows.

But you are prepared now to understand that part of my lesson to come. You see the rings of wood, and the rings of bark, and you want to know how they come there.

Yes, that I do. Of course I could say that the sap makes wood, as I know that bread and butter make flesh; but I want to understand how the change takes place.

Let us then imagine ourselves, Willie, to be spirits able to be inside of the tree, without taking up any room, and with our eyes about us to watch all that is done.

I wish I could do that; it would be better than flying.

Don't talk about it, but jump inside along with me.

I am with you, father, with eyes and ears all open.

I'll tell you what it is, Willie, we shall have not only to be inside the tree, but to be able to stay there a few weeks or months to notice all.

That is easy enough. We may as well fancy hundreds of years as millions of miles.

Let us go to the pith. Around here are vessels in the sheath of the pith. The circles of wood are ranged outside. We will pass along one of the Medullary or pith rays, which you saw make the silver lines of the wood, and here we are at the Liber or inner bark.

But what is this soft white wood next to it?

This is the *Alburnum*, or white wood. When you tap a tree, the sap comes from this part. Now watch what is coming.

I see—the Liber is separating from the soft Alburnum, and there comes a sticking substance of sap oozing between them. What is that?

It is the *Cambium*, or *Albumen*; the white of egg is albumen. Look again.

Ah! the cambium is softening the cellular tissue that bound the liber to the alburnum, and while that is growing outward, I find that the other parts are growing upward.

Let us take a peep at the leaf buds. You see they feel the growing influence, and are letting fall bundles of woody matter, which pass into the cambium, and become attached to the liber. The cellular tissue passes in the mass around, along the medullary rays, and a new substance becomes attached to the liber.

But that new substance is another alburnum, father, and our old alburnum is turned a quite hard wood.

You see, then, that while the last year's soft wood becomes a hard ring or layer, another is formed in its place alongside of the liber. The hard wood is dead compared with the softer kind.

I think I understand that. But where does the nourishment come from?

The sap when first rising is watery and nearly colourless. But after exposure to air and light in the leaf, and losing much water in evaporation, it becomes thicker, and, descending by the bark, is prepared, like blood, to renew and to form different parts of the structure.

Is there any other movement of the fluids of a plant?

Yes, there is a true circulation in the interior of all the cells.

Does a tree form a fresh ring of wood every year?

This depends upon the character of the tree. Some trees take many years in forming it.

How is it that the layers of our Gum tree are not equal in thickness?

That is influenced by the season. One year it was very cold or dry, and the tree made little wood. Another season, the rain was abundant, and the sun did his work well, and so the *concentric* or ring-like layer was fuller.

But I see that on one side of the tree, the bands are thinner than on the other.

That was the side that received the cold wind. The greater the number of leaf-buds above a part of a stem, the greater the diameter.

How does the bark manage when the tree forms new wood, and presses against the liber?

The outer bark cracks and peels off. When the bark grows, it does so internally; a new liber appearing adjoining the layer of new alburnum.

What connection, father, has grafting trees with growing?

The gardener takes a bud of one fruit-tree, and applies its liber, the life part, to the liber of the stock of another tree, binds them together, and puts clay outside to exclude the air. The nature, then, of the bud is communicated to the tree; the juices will be altered, and another fruit produced.

Well, thank you kindly, dear father. I do feel that I know something now about a tree growing. I can tell, too, why a tree does better in one place than another, making more wood and leaves. It is when it can find in the soil a sufficient amount of food, in the carbon, lime, phosphorus, sulphur, potash, and the like.

Quite right, and some other day we may have a talk about the varieties of manures, and the way of tilling the soil, especially in relation to Australian farming. But it is now time for us to return home. So come along, my boy.

It was not until a few days had passed, that Willie's father was at leisure to give another lesson. During that interval the boy had not been idle. He had roamed over the Bush, and stored up lots of specimens of the vegetable world. There were many varieties of roots in one place, and of bark in another. He had cut open many a plant to try to observe some of the peculiarities of which his father had told him. The leaves being so perishable, rather bothered him in his researches, till his mother showed him how to preserve them, by placing them between sheets of blotting or other soft paper, and changing the wrapper occasionally. Willie himself was astonished at his collection, and greatly surprised at the many kinds of leaves he had obtained. Having some notion of drawing, he set about copying a few of the most curious forms, especially in cases where he

thought the pressure would spoil the appearance. His good father, as you may be sure, was quite delighted at his son's industrious research.

One thought had all this while very much perplexed our young friend. He had a fair notion of the causes of the growth of a tree after it had begun to grow; that which puzzled him was, how the tree arose from a seed, and how the seed itself was produced. He unburdened his mind to his father.

To describe this subject, said the father, I must give you the history of a flower.

I used to think, father, that the flowers were only formed to please our eyes, and yet I knew the seeds had something or other to do with them.

You are partly right. There was no reason for flowers to be pretty, any more than for butterflies. But it has pleased God to put the beautiful along with the useful. He has thus adorned this world for our home, that his creatures may be the happier in it. You have doubtless observed that flowers are as various as leaves.

I have done so, and I remarked that the same sort of plant always produced the same kind of blossom, year after year, just as birds lay certain eggs of their own sort only. How is this?

No one can answer that question, except by saying that God has established fixed laws, according to which everything acts in order. The planets turn round the sun, the sun appears every morning; plants and animals produce their like, and everything that God has made moves in its proper path.

Except men and women, father. And yet I think if stars and flowers know and do their duty, we should be at least as good as they.

Now for the flowers, Willie. What do you see in them?

There are the pretty leaves, the common-looking green leaves outside of them, and a lot of things standing upright in the middle of the flower.

Your pretty leaves are the *Corolla* of the flower; the green ones form the *Calyx*; and the standards inside are *Stamens* and *Pistils*. We will take one at a time, and begin with the *Calyx*, although that, and the *Corolla* too, may be wanting in a plant, as they are not really necessary.

It seems as though the Corolla took care of the Stamens, and the Calyx protected the Corolla.

You are not far from the mark, my boy. But can you tell me whether the Corolla of a flower is always in one piece or not?

I should think not. I recollect some flowers with one pretty cup all round, while others are made up of a great many coloured leaves.

You must call those leaves *Petals*. Some plants have four petals in the form of a cross. Did you ever notice a pea blossom?

Yes, father, and thought it a funny one.

Just gather one and examine it. How many petals are there, and how are they placed?

There are five, but not all of the same size. There is a good upper one, two side ones, and two at the bottom, enclosing the rest of the flower.

That upper one is the standard, and the two at the sides are the wings. Could you describe the orchis, the wild flower of our bush?

I will try. There are six petals; three are bent backward, two are at the sides, and one opposite to the three is also bent backward. Is not honey found at the bottom of the corolla cup?

It is, in a place called the *Nectary*. I shall afterwards explain the object of the sweets. But we must hasten on to the *Stamens*. These generally arise from near the base of the petals.

Are they the ones with the yellow dust on the top?

They are, my lad. If you look at one, you will find a long thread or *Filament*, bearing a loose top, called the *Anther*, which carries the *Pollen* or yellow dust.

But why is not the pollen always to be felt on the anther?

Because it is shut up in cells at first. When the walls of these burst open, the pollen shows itself outside, and is also scattered about.

As things alter so strangely under the microscope, father, how would our dust look there?

No two plants have the same shaped pollen, which is of all possible forms. The tiny grains are generally each enclosed in a delicate bag, the subtle powder of which is more like smoke than anything else, in which are particles of starch or drops of oil.

But our orchis does not seem to have a regular anther, for the pollen looks all of a heap.

That is correct. But which do you fancy is the *Pistil*?

I don't know, unless it be that in the middle surrounded by the stamens. This is a thread, too, with a knob at the top.

The support is the *Style*, and the summit is the *Stigma*. When you examine flowers you will observe the stigma of various shapes and sizes; but there is always a sticky substance on the top. Does the pistil rise from the same place as the stamens?

No, it comes from a lump at the bottom.

That lump is the *Ovary*, or seed basket, sometimes known as the *Germen*. Some botanists believe that the several parts of flowers are only metamorphosed, or changed leaves, and that the pistil is one of these growing vertically. You have seen in a double stock, or other double flower, that the stamens have grown into petals.

But why is it that the corolla falls off so soon? I have seen our garden beds strewn with blossoms, and yet the stamens and pistils keep on much longer.

But what happens when these go in their turn, Willie?

Why, the ovary, as you call it, swells out. But what makes the fruit?

That fleshy substance around the seeds, of which you are so fond, is merely the swollen *Pericarp*, or covering of the seed vessel. Children's teeth should not go into the pericarp until the seeds are about ripe.

I know that. When a pear is nice and mellow, then the seeds within are quite hard. But would you call the pod of the pea a pericarp?

Certainly. It consists of two valves, with hinges, and the seed on each side. Sometimes there are several valves together, and we call the whole a *Capsule*. When these burst, the seeds fall out.

But how are the seeds produced?

We are coming to that next. I told you of the bursting of the anther, and the scattering of the pollen. Though much of the dust is wasted, a part is caught on the glutinous stigma of the pistil.

And what becomes of it then?

It passes down the hollow style into the ovary at the bottom.

Yet I have seen flowers in which the pistil was higher than the stamens; how could the pollen reach it?

Many curious contrivances effect the transfer. The wind blows the yellow dust. A bird throws some about. An insect catches some on its wings and legs, and stepping on the stigma, leaves pollen behind.

Is the honey in the nectary to tempt the insects to the flower?

The sweet morsel attracts birds as well as insects.

When, then, the stamens have discharged their pollen, and the pistil conveyed it to the germen, I suppose there is no further use for them.

There is not, and they die off like the petals of the corolla and the calyx before them. But because of the variety of stamens in number and situation, botanists have thought it well to divide plants into classes by this means.

How do they do that, father?

They place in one class all vegetable substances whose flowers contain but one stamen; in another, those with two stamens; and so forth. The number and arrangement of the pistils enable us to form plants into orders; as, a flower may have four stamens, and one pistil, or four stamens and two pistils, &c.

I suppose the stamens and pistils are always found together.

No, they are not: though Compound flowers are made up of sets of flowers within the same calyx, we have plants in which the stamen, or male flowers, are in one part, and the pistil, or female flowers, on another portion of the tree. The former are barren flowers; the latter, fertile or fruitful, as the seed vessel is only connected with the female side.

I think the service of winds, insects, and birds, is more needful than ever now. But I have seen people pick off flowers from melon and cucumber plants, saying, that they were no good.

These were the male blossoms. But they who pluck off all the barren blossoms before the pollen appears, can never expect to have fruit; for the fertile blossoms are of no use without the others. But what would you think of male flowers being all on one tree, while the female are upon another at a distance?

I suppose that is to give the birds and insects more work than ever. Please tell me a tree that has this curious plan.

The Palm is an instance. Where the trees grow in a cluster, as they usually do, there is no difficulty. If very far apart, the male blossoms, when ripe, are cut off, carried to the other tree, and the pollen shaken over the pistil flowers.

That is singular. But how do water plants get on with the yellow dust?

Some of them rise to the surface to flower, and sink to seed. Those that keep below are provided with a glutinous pollen, which cannot be affected with the wet.

Well, father, I do think we are getting on bravely with our lesson. But I would so like to peep into the germen or ovary, and see what goes on there.

We will do so. I ought to have said, that before the pollen sheds, the ovary has a lot of half formed seeds adhering to the *Placenta* inside. These *Ovules*, as they are called, are a sort of nucleus or kernel, and are each placed between two sacs of open mouths.

What goes into those mouths?

Nothing goes in, but the pollen of the stamens comes down through them out upon the ovule, and is received to its *Embryo*, or heart, through a very small hole. Soon after the pollen has entered, the ovule becomes perfected and vivified, or made full of life.

And is that the seed?

The ovules do then become seeds in their little cells, and are, in their turn, shed out when ripe.

But do they turn into real seeds directly the pollen gets down?

By no means. Nourishment and time are necessary to growth and change. I must explain further about the seed. Beneath the *integument* or covering of ordinary seeds, as a pea, what do you observe, Willie?

I have seen the two halves of the seed show themselves, when the skin breaks, and the plant seems to grow from between them.

Your two halves are the *lobes* or *cotyledons* of the seed. These fleshy substances enclose the *embryo*, from which the leaves and roots proceed. The upper part, from which the leaves arise, is the *Plumula*, or *little plume*, or feather. That from which the rootlet proceeds is the *Radicle*, or root part: sometimes it is called the *Rostellum*, or *little beak*.

Of what use are the cotyledons, father?

They supply nourishment to the central portion, which is to develop the plumula and rostellum, and thus surround them. A sort of passage of communication exists between them. There is, also, a neck between the plumula and rostellum.

What is that scar upon the seed?

The point at which it was attached in its vessel.

Well, now I think I really do understand all the mysteries of tree growing. Like as a young animal suckles its parent until strong enough to manage other food, so does a young plant depend upon the cotyledon for provision, until its rootlets are able to search the soil, and leaflets absorb from the air.

I trust, my boy, that while you admire the order and adaptation, you will acknowledge Him whose wisdom planned the whole, whose power performed it, and whose goodness makes it serve the happiness of His creatures.

A few days after the last conversation, Willie went with his father into the Bush among the mountains. The way was long, and the track was rough, but various things occurred to make the journey pleasant. The Laughing Jackass gave the boy a merry greeting, a snake peeped at him slyly among the long grass, and then brushed off quickly to its hole. A Porcupine raised up its quills as he passed, and then trotted off to the scrub. But the trees,—the trees! How tall they were! How thick they were! With the help of their handkerchiefs, Willie and his father measured the girth of some, and found them thirty, forty, and even fifty feet round. Most of the Stringybark and Gum trees ran up straight as a dart, fit to thrust into some monster ship for a mast. Then the sweet-scented shrubs on every side, the green Myrtle, the pyramidal and beautiful Sassafras, and many others, seized upon the boy's imagination, until he came to a place that made him stare, and then caper about like a Black fellow at a corrobory. What could it be?

A break in the tangled brush underwood had given him a peep down into a valley that seemed prettier than anything he had read of in a fairy tale. There was no rough rock, but a floor of soft moss. The most musical of rippling creeks trickled along the vale. No other sound was heard, for the very birds seemed afraid to disturb with their joyous notes the silence that dwelt there. There were no Gums, no Wattles, rising from this moss bed. Instead of these, Willie saw a lot of very odd looking trees. They were not very tall, for they rarely rose above 12 or 15 feet. The bark was unlike anything seen elsewhere. There were no branches, but at the top of the straight stem were several branch-like leaves, 8, 10, and 12 feet long, with such a graceful bend towards the valley. At the top of the stem, rising up from the place where the leaves sprang out, was a curious curling piece, with little shoots. The valley was full of these wonderful vegetable festooned pillars.

Oh! Father! father! cried the boy, what are those beautiful trees down there? I never saw any like them before.

They are Fern trees, my lad. Walk over and look closely at them.

Willie trod upon the soft carpet, and thought it damp, as well as soft. The air, too, was so close, and felt so cold and raw, that he lost half his pleasure already. He could not help saying so to his father.

Ah, my boy, said he, you are not the first to find nice things having something disagreeable about them when approaching nearer to them. Remember, that while roses have thorns, pleasures will have their drawbacks.

Yes, I know I am often very tired after a pretty ramble, and rather queer after a pasty.

True enough; there is only one place where happiness brings neither fatigue nor surfeit. But let us see what is the difference between one of these Fern trees and one of your old companions, a Gum tree.

I can tell. One is like a tree with branches and so forth; but the other is not. Then the bark is quite different. I can see that the leaves of the Fern tree are like those of the common Fern, which is so troublesome in our sandy land, only they are a great deal larger.

But there is a stranger difference still. Just take my axe and cut one of them down.

I can soon do that. It is not nearly so thick as that Wattle I fetched down the other day with not a dozen cuts.

Willie doffs his jacket, catches at the axe, and is at once dropping it in double quick time upon the Fern stem. A dozen blows were given, and another, and another, but the tree did not fall. Taking breath, and wiping his forehead, Willie murmured out:—

Well this is tough! it beats my Wattle, hollow. I declare if the axe don't ring against it as if it were iron or glass. It is quite different from one of our trees, for that is rather soft at the outside, though it may get harder as we get towards the middle.

As I think you are more likely to blunt my axe, than cut down the tree, you may put on your jacket, and we will inspect that fallen fern I see yonder. Perhaps after you have looked at the inside, you will find another wonder.

It was not long before our young Botanist uttered an exclamation of joyful surprise.

I have found it out, said he. Only fancy, father, I was hammering away at a mere shell, though it was a hard one. The tree is quite soft inside, though

like flint outside. It is the very opposite to every tree I ever saw before in all my life.

No doubt it is; and it was to see it with your own eyes that I brought you here to the Fern tree valley.

Thank you, dear father, you always like to give me a treat, and this is a rare one, too. But I am sure that such a tree as this could never grow like a vegetable Gum tree, that you were telling me all about.

Can you name any plants, Willie, of which the fern tree puts you in mind?

I know of none, unless it be the Bamboo, for that is hard outside and hollow within, and it seems, like this, to grow up straight about the same thickness throughout.

There is another tree of this character which we have in Australia, some kinds of which grow in other countries, producing dates, of which boys are so fond.

Any one can guess that:—it is the Palm. But whence have we Palms, father?

There are some in Gipps' Land, at the foot of the Australian Alps; other finer ones are at Illawarra, south of Sydney; but more northerly, in the hotter parts, Palms are very common.

Is that one from which we get our Cabbage Tree Hats?

It is so. The head of the Cabbage Palm is so good tasting and nourishing, that many trees were felled on purpose to get at this sort of cabbage, especially in the early days of the New South Wales Colony.

Then Palms, and Bamboo, and Fern trees grow the same way, as their stems look the same, and as they have neither branches nor solid timber. But how do they grow, father? There is no new wood formed outside the last ring, as in the Gum tree.

It so happens, my boy, that new layers are formed for a while, inside instead of outside. Our new class of trees are, therefore, called *Endogenous*, from their growing internally, in opposition to the ordinary forest trees, which are *Exogenous*, from growing externally.

I suppose the new shoots arise from where we see the joints in the cane; but where is the fruit of our new Endogenous friend?

Have you never seen the pictures of bunches of dates and other fruits all among the leafy branches at the top?

Eh! that I have, with the figures of Blacks climbing up the long poles to get at the fruit.

Now I have two more hard words for you to learn, to distinguish our two sorts of trees. We know them not merely by the difference of wood, but of seed. First tell me how many cotyledons or lobes are in a seed?

There are two in all the seeds that ever I saw.

What! was that cocoa nut the same which you bought the other day?

No, that was all in one piece.

True, Willie; and you may recollect something about a date stone.

I do, indeed; that was single also. I see I must not be so sure another time about what I know. Now I understand. Our new friends of the hard shell stems, which bear fruit at all, have seeds of one piece instead of two.

Quite so. Now a plant of one cotyledon in its seed, is said to be *Monocotyledonous*, while that having two lobes is *Dicotyledonous*; *mono* meaning one, and *di*, two.

Then my Gum tree, and Rose tree, and Sweet Pea, are all *Dicotyledonous*; but the nice Date Palm, the pretty Orchis, and the sweet Cocoa-nut, are *Monocotyledonous*. But are there any plants which are neither one nor the other?

There are some whose seeds are not to be distinguished, and are known as *Acotyledonous*, or *without cotyledons*. Can you not name one?

I don't know, unless you mean such as the Ferns, the Mosses, and the Lichens on the rocks or decaying timber.

They are the very ones. We have now had examples of the three great varieties of vegetation; *Monocotyledonous*, having one lobe to the seed; *Dicotyledonous*, having two lobes; and *Acotyledonous*, without any.

Will you now, dear father, explain the way in which the Monocotyledonous plants differ from the Dicotyledonous, especially in their system of growth?

I will, my boy, and we will begin this time with the seed.

Any one can tell where the plumula and rostellum of the Cocoa-nut proceed, for we can see the two places in the shell, one at the top and the other at the bottom. But do tell me what the white of the Cocoa-nut is?

That sweet substance is the *Albumen*, to nourish the young embryo inside. In most Monocotyledonous seeds, there is a lot of this albumen—which is thus chemically changed to suit for food. In the Date Palm this store-house is very hard, as you well know.

Is the Embryo different from others?

In general it is like an undivided cylinder, but in Grass and Corn, it is a flat plate.

What! is Corn like the Palm, father?

Yes; for although it has a second, unshapen lobe, it is truly monocotyledonous.

Well, father, how does the growth of the Mono folks go on?

In the Palm embryo are noticed bundles of fine tissue, arranged circularly from the plumula to the radicle. These tissues increase and shoot downwards. The cellular tissues of the embryo now enlarge to make room for woody fibres, which begin to appear. A rootlet is formed, and a leaf comes forth.

But how is the stem formed?

There are several crown leaves springing from the root in a circular form, each circle with its own cluster of vessels. Each leaf bud sends out a lot of woody bundles, curving downwards and outwards, which become entangled irregularly in the cellular tissue.

That is how it is that there seems no order and arrangement about the wood of a Palm or Fern tree. All is higgledy-piggledy, and the moist stuff must be collected near the circumference. But what is the soft centre made up of?

Cellular tissue, of course. You can distinguish no pith, no medullary rays, and no distinct bark, like our fruit trees.

No, I see that which ought to be the bark is a jumble of fibres, which cannot be separated from the stem as in the regular trees. I don't like these monocotyledonous trees; they seem quite ungainly and unnatural.

And yet you have no objection to the bread of corn, the sugar of the cane, the nut of the cocoa, the date of the palm, the banana, the yam, the arrow-root——

Stop, stop, father, you have given such a list of good things, that I will never say a word against the monocotyledonous family as long as I live, though I do wish the botanists would give them a name that boys could pronounce better.

Something must be said about the leaves of these curious plants. Instead of their veins being branched out, the leading ones run parallel, and are connected by small cross veins. For this reason an injury to any part of a palm leaf is of much more consequence than if it happened to a gum leaf. You have seen that the leaves are the true organs by which woody matter is formed, and not the inner bark, as with other trees.

Is there any peculiarity about the flowers of Endogenous trees?

Their petals, stamens, and pistils, are generally in threes or sixes. The palm has usually six stamens. Half the plants of New Zealand are monocotyledonous, being far more than in Australia.

Will you please, father, say a few words about the third sort of plants—the *Acotyledonous*?

These are cellular, and have no apparent flowers.

But how do they propagate or spread themselves, if they have no regular seeds?

They have little things called *Sporules*, or *Spores*, without spiral vessels, which are the means of increase. Mosses have capsules, with teeth, as

before mentioned; the fringe enclosing a rude kind of flower. Some of these are barren, and others fertile. The latter may have several imperfect pistils along with one perfect one.

I have half a mind, father, to fancy that the common fern is one of the acotyledonous.

Some botanists think the same. The seeds are formed upon the back and margin of the fronds or leaves, in the form of bud-like spores. They are exceedingly small, and no flower is visible to the eye.

Do these spores, like seeds, produce stem from one part, and root from the other?

No, my boy, the little globular things have no plumula and radicle; but that part which is exposed to the light produces stem, and from the shaded portion the root descends.

Are other plants produced from spores?

Yes, Willie, the mushrooms and sea-weed are. The productive organs of lichens, the most imperfect of plants, are the *Shields*, as they are called, upon the *Thallus*, or leaf-stem.

Well, father, I think now we have gone over all the different sorts of plants and trees, and I feel myself the wiser for your kind and pretty lessons. I shall be a regular botanist now.

Not so fast, my little man; you have only begun your lesson. There is plenty more to learn. There are many thousands of names of plants, most of which are hard to understand, and many are hard to pronounce. Then you have to know the arrangements of all these, and their peculiar natures and uses.

Ah! I see my work is not done. Never mind, you have taught me the alphabet of botany. I mean now to study all the flowers and trees I come to, with the help of your letters, until I can read off easily all about their peculiar habits and modes of growth.

Do so, my son; but don't, like some philosophers, elbow God out of the field. Be ready to notice His thoughtfulness and goodness in the world of plants; and, as they in a dumb manner show forth His praise, be you willing, with heart and voice, to say, "My Father made them all."

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